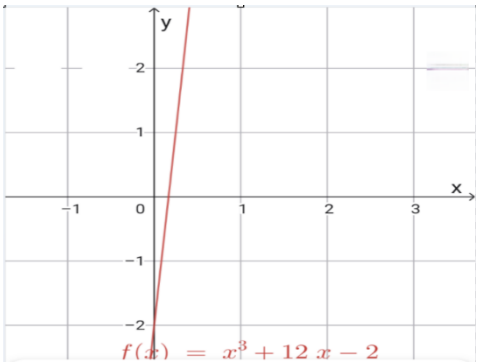
	FURTHER MATHEMATICS TERM 1 – EXAM N° 1 – TIME ALLOWED: 03 HR 00 MINS – COEF: 5 DATE: OCTOBER 2026 - SCHOOL YEAR: 2026/2027	TOTAL MARK OF ALL SECTIONS	FINAL SCORE / 20
CANDIDATE'S NAME		A:	
CLASS	UPPER SIXTH SCIENCE	B:	
TEACHER'S NAME	NDOKMO Hermann, TUKOV Louis	Total:	

1. The graph below is that of the function $f(x) = x^3 + 12x - 2$.



a. State the variation of f **(1 mark)**

b. Justify that $f(x) = 0$ has only one root α(not to be calculated) and that $0 < \alpha < 1$

(2 marks)

c. Study the sign of f **(2 marks)**

Let g be a real valued function defined by $g(x) = \frac{x^3+1}{x^2+4}$.

d. Show that g is differentiable in R and that $g'(x) = \frac{xf(x)}{(x^2+4)^2}$ **(3 marks)**

e. Study the variation of g **(2 marks)**

f. Proof that $g(\alpha) = \frac{3-12\alpha}{\alpha^2+4}$ **(2 marks)**

g. Show that $-\frac{9}{5} < g(\alpha) < \frac{3}{4}$ **(2 marks)**

h. Show that $g(x)$ can be expressed as $x - 4 + \frac{k}{x^2+4}$, where k is a constant to be determined. Hence state the oblique asymptote of g **(3 marks)**

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i. Study the relative position of g and its oblique asymptote **(1 mark)**

j. Sketch the graph of y = g(x) showing clearly its behaviour as it approaches its asymptote **(2 marks)**

2. (a)Find all natural numbers n such that their squares is a factor of 40425 **(2 marks)**

Let x and y be two positive integers such that gcd (x,y) = d and lcm(x,y) = m.

b. Write a relationship between d and m **(1 mark)**

Given also that $m^2 - 3d^2 = 40425$

c. Find the values of x and y. **(4 marks)**

3. Let $f(x) = \sqrt{x^{-2}+8} - x$.

a. Show that f(x) can be expressed as $f(x) = \frac{8}{\sqrt{x^2+8}+x}$ **(1 mark)**

b. Prove that $\forall x \in [1,2]$, $1 \leq f(x) \leq 2$ **(2 marks)**

c. Show equally that $\forall x \in [1,2]$, $|f'(x)| \leq \frac{2}{3}$ **(3 marks)**

Let (U_n) be a sequence defined by $\begin{cases} U_0 = 1 \\ U_{n+1} = f(U_n) \end{cases}$

d. Show that for all natural number n, $1 \leq U_n \leq 2$. **(3 marks)**

e. By using the mean value theorem, show that for all $n \in N$,

$$\left|U_{n+1} - 2\sqrt{\frac{2}{3}}\right| \leq \frac{2}{3}\left|U_{n+1} - 2\sqrt{\frac{2}{3}}\right| \quad \textbf{(2 marks)}$$

f. Show that $\left|U_n - 2\sqrt{\frac{2}{3}}\right| \leq \left(\frac{2}{3}\right)^n$ **(2 marks)**

g. Deduce that (U_n) converges towards a real number l to be determined. **(2 marks)**

4. Let $f: R \longrightarrow R$

$$x \longmapsto \frac{x+1}{\sqrt{x^2+1}} - 1$$

a. Study the variation of f **(2 marks)**

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Let (C) be the curve representative of f

b. Find the equation of the tangent (T) of (C) at the point x = 0 **(2 marks)**

c. Study the relative position of (T) and (C) **(2 marks)**

d. Sketch (C) **(2 marks)**

Let's consider the sequence (U_n) defined by $\begin{cases} U_0=-\frac{1}{2} \\ U_{n+1}=f(U_n) \end{cases}$

e. Prove that $\forall n \in N, -1 < U_n < 0$ **(3 marks)**

f. Show that U_n converges **(2 marks)**

g. Prove that $\forall n \in N, 0< U_{n+1} + 1 \leq \frac{2\sqrt{5}}{5}(U_n + 1)$ **(3 marks)**

h. Show equally that $\forall n \in N, 0 < U_n + 1 \leq \frac{1}{2}\left(\frac{2\sqrt{5}}{5}\right)^n$ **(2 marks)**

i. Hence find the limit of U_n **(2 marks)**

5. Given that m and n are positive integers and that

$$I_{m,n} = \int_1^e \frac{(\ln x)^m}{x^n} dx$$

i. Show that $(n-1)I_{m,n} = -\frac{1}{e^{n-1}} + mI_{m-1,n}$ **(3 marks)**

ii. Hence, find the volume of the solid generated when the curve y = f(x), where f(x) where f(x) = $\frac{\ln x}{x}$ fromx = 1 to x = e is rotated completely about the x- axis **(4 marks)**

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6. Three forces F_1, F_2 and F_3 with position vectors r_1, r_2 and r_3 respectively, act on a body P of mass 4kg. Given that:

$F_1 = (i+2j-3k)N$ $r_1 = (2i-j+2k)m$

$F_2 = (2i-j+5k)N$ $r_2 = (-i+3j-k)m$

$F_3 = (-i+j+k)N$ $r_3 = (3i+j-k)m$

(a) Determine the resultant force on P and deduce its acceleration
(b) Find the resultant moment on P
A fourth force F_4 with position vector r_4 = (ai+2j-k)m, together with the above three

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forces form a couple of magnitude $\sqrt{341}$Nm, about the origin.

(c) Find the value(s) of t.

(d) If the force F_2 displaces a particle from the point A(2,1, -3) to the point B(5, -1, 1), find the work it does. **(3, 4, 3, 3 marks)**

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7. (i)

(a) Given that y satisfies the differential equation $\frac{dy}{dx} - xy = y^2$ and that y = 2 when x = 0. Find the MacLaurin's expansion of the function y, upto and including the term in x^4.

(b) Use Simpson's rule and six ordinates to find an approximate value of the area bounded by the curve of the function $y = \frac{1}{2+5x}$, the x-axis and the lines x = 1 and x = 3 leaving your answer as a fraction.

(ii) Given that y satisfies the differential equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + xy = 0$, use the approximations $\left(\frac{d^2y}{dx^2}\right)_n \approx \frac{y_{n+1} - 2y_n + y_{n-1}}{h^2}$ and $\left(\frac{dy}{dx}\right)_n \approx \frac{y_{n+1} - y_{n-1}}{2h}$ to show that

$$(2-3h)y_{n+1} \approx 2(2-h^2x_n)y_n - (3h + 2)y_{n-1}$$

Using a step length of h = 0.15, find correct to three decimal places, the value of y when x = 0.6 given that y = 1 at x = 0 and at x = 0.15 **(5, 4, 3, 6 marks)**

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